

RNO parameters

Signal Strength Unit

This is the unit that is used when signal strength is presented in the graphical user interface. In FAS/FOX the measured signal strength is actually measured interference.

Type: String

Value range: "dBm" or "GSM"

Installation setting: dBm

Used by: FAS, FOX, NCS, TET, MRR

Co-Frequency Distance (km)

This parameter is used for two things:

1. It determines the furthest distance to look for co-channel frequency cells affected by change flags in the FAS Overview Comparison Report.
2. It determines the furthest distance to look for cells when creating cell sets with cells that have the same frequency allocated as the selected cell.

Type: Integer

Value range: 0 - 100 km

Installation setting: 20 km

Used by: FAS, Set Handling - New Cell Set

Adjacent Frequency Distance (km)

This parameter is used for two things:

1. It determines the furthest distance to look for adjacent-channel frequency cells affected by change flags in the FAS Overview Comparison Report.
2. It is also used when creating new cell sets. It determines the furthest distance to look for cells which have an allocated frequency that is an adjacent frequency to an allocated frequency in the selected cell.

Type: Integer

Value range: 0 - 100 km

Installation setting: 10 km

Used by: FAS, Set Handling - New Cell Set

No. of Test Frequencies to Add at Each Interval

This is the default value for the "No. of Test Frequencies to Add at Each Interval" entry field in the FAS/FOX, NCS/NOX and TET - New Recording dialog box. It is the default number of frequencies that may be added to the active BA-list at each interval.

Type: Integer

Value range: 0 - 32

Installation setting: 5

Used by: FAS, FOX, NCS, NOX, TET

Pathloss per Distance Decade (dB/decade)

This parameter is used when calculating probable neighbour cell identities from measured BCCH/BSIC pairs. It specifies the decrease in signal strength (dB) when distance becomes 10 times greater.

Type: Integer

Value range: 0 - ...

Installation default: 31 dB/decade

Used by: FAS, FOX, NCS, NOX

Sector Antenna Gain Offset (dB)

This parameter is used when finding probable neighbour cell identities. It specifies the average (over all directions) antenna gain for a sector antenna relative to an omni antenna, measured in dB. For example, say sector antennas amplify 18 dB forwards and -10 dB backwards, and that omni antennas amplify 9 dB. This gives an offset of $(18 + -10) / 2 - 9$, that is -5 dB.

Type: Integer

Value range: ... - ...

Installation default: -5 dB

Used by: FAS, FOX, NCS, NOX

Sector Antenna Gain Amplitude (dB)

This parameter is used when finding probable neighbour cell identities. It defines the amplitude of the direction variation for a sector antenna, in dB. Antenna gain for sector antennas are modeled as a cosine function with this value as amplitude. For example, say sector antennas amplify 18 dB forwards and -10 dB backwards, then $(18, -10)$, the value is $(18 - -10) / 2$, that is 14 dB.

Type: Integer

Value range: ... - ...

Installation default: 14 dB

Used by: FAS, FOX, NCS, NOX

Weak Probable Cell Name Choice Warning Threshold (dB)

This parameter is a threshold used to warn for insufficient BSIC planning in FAS/FOX. To identify which cell a certain BSIC/BCCH combination resembles, RNO calculates a predicted signal strength from all cells with matching BCCH/BSIC. It then chooses the strongest of these cell as the probable (interfering) cell. If the difference between the predicted signal strength of the chosen cell and the second best cell is lower than this value, a warning will be issued to the recording log.

Type: Integer

Value range: 0 - ...

Installation default: 5 dB

Used by: FAS, FOX, NCS, NOX

BA List Change Interval (minutes)

This is the default value for the "BA List Change Interval" entry field in the FAS/FOX, NCS/NOX and TET - New Recording dialog box. It is the default

time, in minutes, a test frequency should be in the active BA-list.

Type: Integer

Value range: 10 - 1440 minutes

Installation setting: 10

Used by: FAS, FOX, NCS, NOX, TET

Network Update Time

This is the time of the day when changes to the network determined from RNO expert recordings shall be performed. It is specified as hour and minute separated by a colon.

Type: String

Installation setting: 02:45

Used by: FOX, NOX

ounding Box (km)

This defines the distance (in km) by which the smallest possible rectangle, enclosing all cells in the cell set is stretched, to obtain the "bounding box". The bounding box area plus 1st and 2nd order neighbours includes all cells that are used in the process of identifying the measured BAR file neighbouring cell data records as originating from a particular cell.

Type: Integer

Value range: 0 - ... km

Installation setting: 50 km

Used by: FAS, FOX, NCS, NOX

Pathloss per Distance Decade (dB/decade)

This parameter is used when calculating probable neighbour cell identities from measured BCCH/BSIC pairs. It specifies the decrease in signal strength (dB) when distance becomes 10 times greater.

Type: Integer

Value range: 0 - ...

Installation default: 31 dB/decade

Used by: FAS, FOX, NCS, NOX